NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 2

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total marks: 10 x 1 = 10**

QUESTION 1:

An IP packet arrives with the first eight bits as 0100 0101. How many bytes are there in the OPTIONS field?

Correct Answer: 0

Detail Solution: First four bits (0100) = version 4. Next four bits (0101 = 5) = header length. Header length = 5 x 4 = 20 bytes, which is the minimum IP header size with no options. Thus the OPTIONS field size = 20 - 20 = 0 bytes.

QUESTION 2:

Which of the following is/are true for TCP header flag bits?

a. SYN=1, ACK=0 indicates a connection request.
b. SYN=1, ACK=1 indicates connection confirmation.
c. FIN flag is used to reject a connection.
d. RST flag is used to gracefully terminate a connection.
e. URG flag indicates urgent data in the segment.

Correct Answer: a, b, e

Detail Solution: SYN=1, ACK=0 is a connection request; SYN=1, ACK=1 is connection confirmation; URG flag indicates urgent data. FIN is used for graceful connection termination (not rejection), and RST is used to abruptly reset/reject a connection (not graceful termination). Thus, the correct options are (a), (b), and (e).

QUESTION 3:

Which of the following represents a valid broadcast address for its respective subnet (as specified)?

a. 10.255.255.255 for network 10.0.0.0/8
b. 10.20.0.0 for network 10.20.0.0/16
c. 192.168.5.255 for network 192.168.5.0/24
d. 172.16.255.0 for network 172.16.0.0/16
e. 192.168.5.0 for network 192.168.5.0/24

Correct Answer: a, c

Detail Solution: For 10.0.0.0/8, the host portion is 24 bits, so broadcast address = 10.255.255.255 (all host bits set to 1). For 192.168.5.0/24, the host portion is 8 bits, so broadcast address = 192.168.5.255. Option (b) is the network address, not broadcast. Option (d) is incorrect since for /16, broadcast address should be 172.16.255.255, not 172.16.255.0. Option (e) is the network address, not a broadcast address.
Thus, the correct options are (a) and (c).

QUESTION 4:

An IP packet arrives at the final destination with D flag = 0, M flag = 0, and offset = 5. Which of the following is true?

a. The packet has not been fragmented.
b. The packet is the first fragment.
c. The packet is the last fragment of a fragmented packet.
d. The packet will be dropped due to invalid flags

Correct Answer: c

Detail Solution: M = 0 indicates no more fragments follow, meaning this is the last fragment. A non-zero offset (5) confirms fragmentation has occurred, ruling out option (a). Hence, it is the last fragment of a fragmented packet.
Thus, the correct option is (c).

QUESTION 5:

Which of the following IP address ranges are reserved for private use?

a. 10.0.0.0 to 10.255.255.255
b. 100.64.0.0 to 100.127.255.255
c. 172.16.0.0 to 172.31.255.255
d. 192.168.0.0 to 192.168.255.255
e. 169.254.0.0 to 169.254.255.255
f. 224.0.0.0 to 239.255.255.255

Correct Answer: a, c, d

Detail Solution: The private IP ranges as per RFC 1918 are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16. 169.254.x.x is reserved for link-local addressing (APIPA), 100.64 to 100.127 is for CGNET, and 224.0.0.0 to 239.255.255.255 is the Class D multicast range.
Thus, the correct options are (a), (c), and (d).

QUESTION 6:

Which of the following well-known port number(s) is/are correctly matched with their corresponding service?

a. HTTP – 80
b. FTP – 21
c. SSH – 25
d. DNS – 23
e. Telnet – 53
f. SMTP – 22

Correct Answer: a, b

Detail Solution: HTTP uses port 80, FTP uses port 21, DNS uses port 53, and Telnet uses port 23, SSH uses port 22, and SMTP uses port 25.

Thus the correct options are (a) and (b).

QUESTION 7:

What is the subnet mask corresponding to CIDR notation /26?

a. 255.255.255.0
b. 255.255.255.128
c. 255.255.255.192
d. 255.255.255.224
e. 255.255.255.240

Correct Answer: c

Detail Solution: /26 means 26 network bits, leaving 6 host bits. The last octet becomes 11000000 = 192. So the subnet mask is 255.255.255.192.

Thus the correct option is (c).

QUESTION 8:

Which of the following statement(s) is/are true about IPv4 fragmentation and the "Don't Fragment" (DF) bit?

a. If DF = 1 and packet size exceeds MTU, the packet is dropped and an error is generated.
b. If DF = 1, the router fragments the packet regardless of size.
c. If DF = 0, fragmentation is allowed if necessary.
d. DF bit is part of the Fragment Offset field.
e. DF = 1 guarantees the packet will always reach its destination.

Correct Answer: a, c

Detail Solution: If DF = 1 and the packet exceeds the outgoing MTU, the router drops the packet and typically sends an ICMP "Fragmentation Needed" error. If DF = 0, fragmentation is permitted when required. DF is part of the Flags field, not Fragment Offset, and setting DF=1 does not guarantee delivery.

Thus the correct options are (a) and (c).

QUESTION 9:

An IP packet has HLEN = 12 and TOTAL LENGTH = 900 bytes. The number of data bytes in the packet will be _____.

Correct Answer: 852

Detail Solution: HLEN = 12 means header size = 12 x 4 = 48 bytes. Data bytes = 900 - 48 = 852 bytes.

Thus the correct answer is 852.

QUESTION 10:

_____ is the size (in bytes) of a UDP header.

Correct Answer: 8

Detail Solution: The UDP header has 4 fields (source port, destination port, length, checksum), each of 2 bytes, totaling 8 bytes.
Thus, the correct answer is 8.

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